Mobile and Autonomous Robots (UE23CS343BB7)

Jackfruit Problem

(Mini-Project Report)

Project Title: Space exploration robot

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1. Introduction

The Space Exploration Robot project demonstrates a planetary rover in a ROS 2 and Gazebo simulation environment. The core motivation is to test rover mobility and controllability in rough extraterrestrial-like terrain before moving to costly physical prototypes. In such terrain, wheel slip, uneven contact, and directional stability become important challenges. Our implementation addresses this by combining a six-wheel rover model, steering and velocity control configuration, sensor integration, and manual teleoperation for controlled testing.

2. Objective

- Develop a simulation-ready space rover model suitable for moon-surface testing.
- Configure wheel and steering controllers to enable stable movement and directional control.
- Integrate and validate key onboard sensors required for robotic exploration workflows.
- Demonstrate manual operation using keyboard teleop and observe rover response in real time.
- Evaluate rover behavior on uneven lunar terrain, including turning and traversal stability.

Expected outcomes: Smooth command-to-motion behavior, clear sensor setup, and robust rover traversal over irregular moon terrain in simulation.

Project scope: rover design, model configuration, sensors, keyboard teleoperation, and moon terrain movement demonstration.

3. Methodology

We followed a layered approach. First, the rover structure was defined through URDF/Xacro along with control interfaces. Second, controller configuration was applied for steering and wheel velocity control. Third, moon terrain simulation was launched in Gazebo with lunar gravity settings. Finally, teleop commands were applied and rover response was observed to verify practical movement on uneven surfaces.

3.1 Algorithms

- **Keyboard command mapping:** Key presses are translated into linear and angular velocity commands published on cmd_vel.
- **Velocity limiting and normalization:** Incoming velocities are bounded and normalized to control-friendly ranges.
- **Wheel velocity distribution:** During turns, separate wheel speeds are computed using rover geometry so inner and outer wheels move proportionally.
- **Steering target computation:** Corner steering angles are computed from turning radius and mapped to actuator command ticks.
- **Execution flow:** User key input → cmd_vel message → parser node → per-wheel and steering command generation → rover actuation in Gazebo.

3.2 Tools Used

- **ROS 2 (Humble):** Middleware, nodes, topics, and launch orchestration.
- **Gazebo:** Physics-based simulation of moon terrain and rover dynamics.
- **URDF/Xacro:** Robot body, wheel assembly, and joint configuration modeling.
- **ros2_control:** Position control for steering joints and velocity control for drive wheels.
- **Python/C++ packages:** Teleop, parsing, and controller integration modules.
- **Sensors configured:** Hokuyo LiDAR, IMU, and RGB-D camera in simulation.

4. Outcome

The rover performed successfully in moon terrain simulation. We validated robot model setup, controller response, and sensor configuration. During keyboard teleoperation, the rover handled forward/reverse motion and turns while maintaining stable posture over uneven terrain patches.

4.1 Result Screenshots

Figure 1: Rover deployed in moon simulation environment with full model and terrain context.

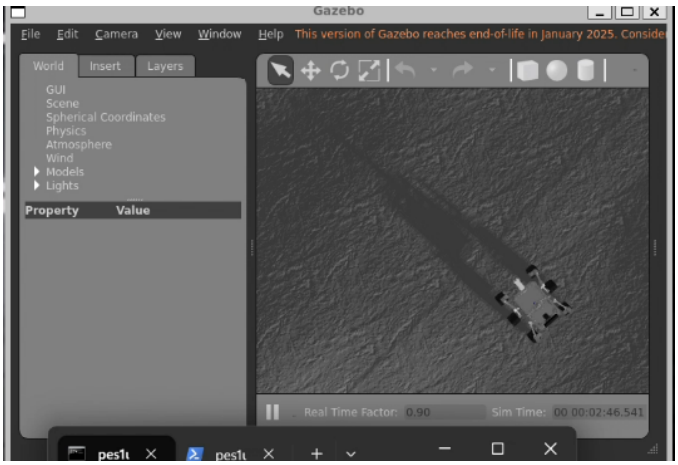


Figure 2: Teleoperation-driven rover motion showing direction and control response.

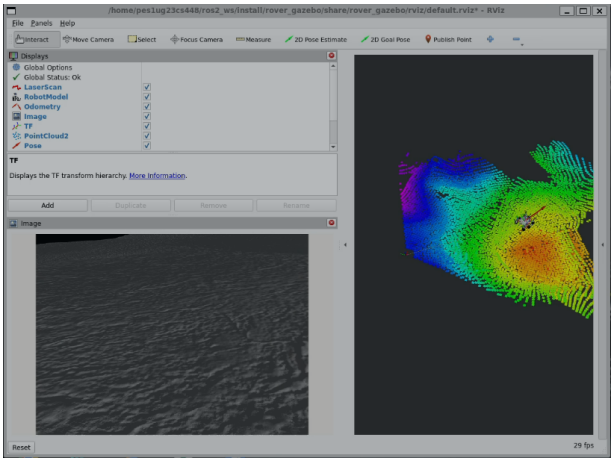
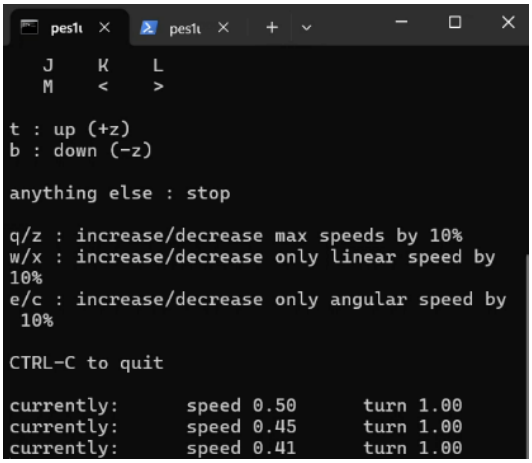


Figure 3: Rover traversing uneven region, showing suspension and wheel-ground adaptation.



4.2 Video Demonstration

Drive Link: https://drive.google.com/file/d/1lscbfopLCT67udzZggwgmZdnvsF6yqPD/view?usp=sharing

The video demonstrates rover design, configuration, sensors, keyboard teleoperation, and movement over moon terrain.

4.3 GitHub Repository

GitHub Link: https://github.com/Pratheek22/Space-Rover

Repository contains the source code, robot description files, launch setup, and simulation assets used for this project.

5. Conclusion

- **Key learnings:** Practical integration of robot modeling, ROS 2 communication, controller tuning, and simulation-based validation.
- **Challenges faced:** Command tuning for smooth turning, maintaining consistent behavior on irregular terrain, and coordinating multiple package configurations.
- **Final achievement:** A complete and demonstrable moon-terrain rover simulation controlled through keyboard teleoperation with validated sensor and control setup.